

Funnel Analysis Documentation

1. Project Overview

The Funnel Analysis Dashboard is designed to analyze the customer journey from browsing a product to completing a purchase. The dashboard helps identify how users move through different stages of the sales funnel and highlights where users are dropping off during the journey.

The analysis provides an overview of user activity across the Browse, Add to Cart, Checkout, and Purchase stages. It also allows users to analyze funnel performance based on channel, product category, region, and device.

2. Project Objective

The main objectives of this project are:

- To analyze the customer journey through different funnel stages.
- To measure the number of users at each stage of the funnel.
- To calculate conversion rates between different stages.
- To identify cart and checkout abandonment.
- To compare user activity across different marketing channels.
- To analyze customer behavior across product categories, regions, and devices.
- To identify areas where the business can improve its conversion rate.
- To provide an interactive dashboard for monitoring funnel performance.

3. Tools Used

- **Power BI** – Dashboard development and data visualization
- **Power Query** – Data preparation and transformation
- **DAX** – Calculations and performance metrics

4. Dataset Description

The Funnel Analysis dataset contains information about user interactions during their online purchasing journey. The dataset is used to understand how users move from the initial browsing stage to the final purchase stage.

The main funnel stages considered in the analysis are **Browse, Add to Cart, Checkout, and Purchase**. Along with these stages, the dataset contains additional information such as user ID, session ID, event time, channel, device, region, product category, revenue, and other relevant fields.

This dataset helps in understanding customer behavior and identifying the stages where users are successfully moving forward and the stages where a significant number of users are dropping off.

5. Data Preparation and Transformation

Before creating the dashboard, the dataset was reviewed and prepared for analysis. Data preparation is an important step because the accuracy of the dashboard depends on the quality and consistency of the underlying data.

The following steps were performed during the data preparation process:

5.1 Data Inspection

The dataset was first examined to understand the available columns, data types, funnel stages, and other attributes. The required fields for funnel analysis were identified and reviewed.

5.2 Data Cleaning

The data was checked for missing values, incorrect data types, duplicate or inconsistent records, and other possible data-quality issues. The data was cleaned and organized so that it could be used effectively for analysis.

5.3 Funnel Stage Preparation

The different customer journey stages were organized into the required funnel sequence:

Browse → Add to Cart → Checkout → Purchase

This sequence helps to measure how users progress through the purchasing journey and where they are lost before completing a purchase.

5.4 Creating Measures

Required measures were created to calculate important business metrics such as total revenue, number of browse users, add-to-cart users, checkout users, purchase users, conversion rates, and abandonment rates.

These measures were then used in the KPI cards and other dashboard visualizations.

5.5 Creating Interactive Filters

Slicers were added to make the dashboard interactive. The dashboard includes filters for **Stage, Region, Event Time, Channel, and Device**.

These filters allow users to select a particular category and analyze funnel performance based on the selected criteria.

6. Key Performance Indicators

Several KPIs were created to provide a quick overview of the overall funnel performance.

Total Revenue

The Total Revenue KPI shows the revenue generated from completed purchases. In the dashboard, the total revenue is displayed as approximately **277.32K**.

Browse Users

This KPI represents the number of users who entered the funnel by browsing products. The dashboard shows approximately **10K browse users**.

Add to Cart Users

This KPI represents the users who added products to their shopping cart after browsing. The dashboard shows approximately **7K users** at this stage.

Checkout Users

This KPI represents users who proceeded from the cart to the checkout stage. The dashboard shows approximately **3K checkout users**.

Purchase Users

This KPI represents users who successfully completed their purchase. The dashboard shows approximately **1K purchase users**.

Browse to Cart Rate

The Browse to Cart Rate measures the percentage of browse users who proceeded to add products to their cart. The dashboard shows a rate of approximately **69.49%**.

Purchase Rate

The Purchase Rate indicates the percentage of users who successfully completed the purchase process. The dashboard shows a purchase rate of approximately **29.05%**.

Overall Conversion Rate

The Overall Conversion Rate measures the overall percentage of users who moved through the funnel and completed a purchase. The dashboard shows an overall conversion rate of approximately **10.04%**.

Cart Abandonment Rate

The Cart Abandonment Rate indicates the percentage of users who added products to their cart but did not successfully complete the next stage of the purchase journey. The dashboard shows a cart abandonment rate of approximately **85.55%**.

Checkout Abandonment Rate

The Checkout Abandonment Rate measures the percentage of users who reached the checkout stage but did not complete the purchase. The dashboard shows a checkout abandonment rate of approximately **70.95%**.

7. Dashboard Development and Visualizations

After completing the data preparation and creating the required measures, an interactive Funnel Analysis Dashboard was developed using Power BI.

The dashboard was designed to provide a clear view of customer movement through the different stages of the purchasing journey. KPI cards were placed at the top of the dashboard to display the most important performance metrics.

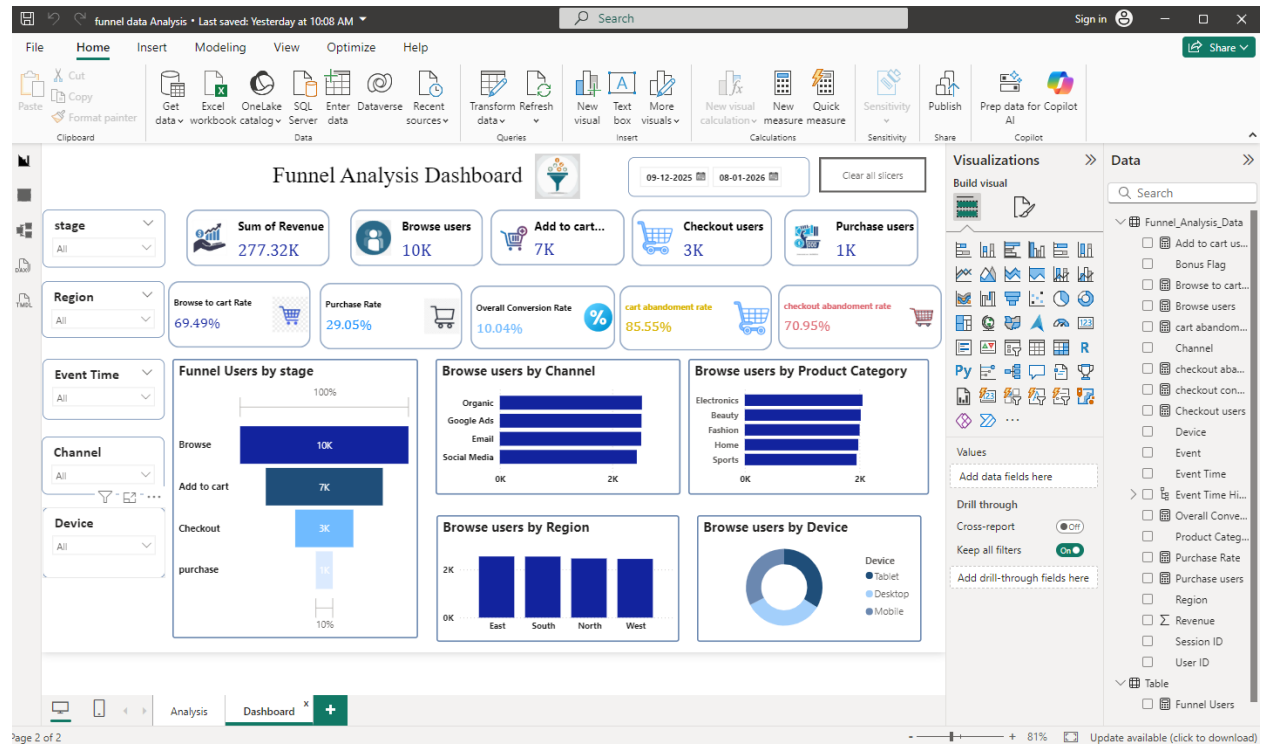
A funnel chart was created to show the number of users at each stage of the customer journey. The funnel starts with Browse users and gradually reduces through Add to Cart, Checkout, and Purchase stages. This makes it easier to identify the drop-off between stages.

Additional visualizations were created to analyze browse users based on different business dimensions:

- **Browse Users by Channel** – compares user activity across Organic, Google Ads, Email, and Social Media channels.
- **Browse Users by Product Category** – compares browsing activity across categories such as Electronics, Beauty, Fashion, Home, and Sports.
- **Browse Users by Region** – compares the number of browse users across East, South, North, and West regions.
- **Browse Users by Device** – shows the distribution of users across devices such as Tablet, Desktop, and Mobile.

The dashboard also contains slicers for **Stage, Region, Event Time, Channel, and Device**. These slicers allow users to interact with the dashboard and analyze specific segments of the customer journey.

Dashboard Screenshot



The completed dashboard provides an interactive and visual representation of the customer funnel, making it easier to monitor conversion performance, identify customer drop-offs, and understand user behavior across different segments.

8. Key Insights

The Funnel Analysis Dashboard provides several important insights into customer behavior and the overall purchasing journey.

8.1 Customer Drop-Off Across Funnel Stages

The number of users decreases significantly as they move from Browse to Add to Cart, Checkout, and finally Purchase. While approximately **10K users** entered the funnel at the Browse stage, only around **1K users** completed the purchase.

This indicates that a large proportion of users are lost before reaching the final purchase stage. Therefore, improving the conversion process at the later stages of the funnel can help increase the number of completed purchases.

8.2 High Cart Abandonment

The dashboard shows a **Cart Abandonment Rate of approximately 85.55%**. This indicates that a significant number of users who add products to their cart do not successfully proceed through the purchasing process.

This could indicate potential issues such as complicated checkout procedures, unexpected costs, lack of preferred payment options, or customers comparing products before making a final decision.

8.3 Checkout Abandonment

The **Checkout Abandonment Rate is approximately 70.95%**, which indicates another major drop-off point in the customer journey.

Businesses can investigate the checkout experience to understand why users leave before completing their purchases. Simplifying the checkout process and providing a smoother payment experience may help improve conversion.

8.4 Channel Performance

The dashboard compares browse users across different channels, including **Organic, Google Ads, Email, and Social Media**.

This comparison helps identify which channels are bringing more users into the funnel. However, the channel generating the highest number of users should not automatically be considered the best-performing channel. Businesses should also compare the conversion and purchase performance of users coming from each channel.

8.5 Product Category Analysis

The dashboard provides a comparison of browse users across different product categories such as **Electronics, Beauty, Fashion, Home, and Sports**.

This helps identify categories that attract higher customer interest. Businesses can use this information to understand customer preferences and plan suitable marketing and promotional strategies.

8.6 Regional Analysis

The dashboard compares user activity across different regions, including **East, South, North, and West**.

Regional analysis helps businesses identify areas with higher customer engagement and areas where additional marketing efforts may be required.

8.7 Device Analysis

The dashboard also analyzes users based on the device they use, such as **Mobile, Desktop, and Tablet**.

Understanding device-wise customer behavior can help businesses optimize their website and purchasing experience for the devices used most frequently by customers.

9. Business Recommendations

Based on the funnel analysis, the following recommendations can be considered:

- Simplify the checkout process to reduce customer drop-off.
- Investigate the reasons for high cart abandonment.
- Provide multiple convenient payment options.
- Make additional charges and delivery costs clear before checkout.
- Optimize the purchasing experience for mobile and other frequently used devices.
- Analyze conversion performance by marketing channel instead of focusing only on traffic volume.
- Provide targeted offers and promotions for high-performing product categories.
- Monitor regional performance and focus marketing efforts on regions with growth potential.
- Continuously track funnel conversion rates to identify changes in customer behavior.

10. Conclusion

The Funnel Analysis Dashboard provides a clear view of the complete customer purchasing journey, from browsing products to completing a purchase. The analysis helps identify major customer drop-off points and provides insights into channel, product category, region, and device performance.

The dashboard can help businesses make data-driven decisions to improve customer experience, reduce abandonment, increase conversion rates, and ultimately improve revenue.

Overall, the project demonstrates the use of **data cleaning, data transformation, DAX measures, KPI analysis, interactive filters, and Power BI visualizations** to convert raw customer data into meaningful business insights.